

Project Initialization and Planning Phase

Date	20 September 2026
Team ID	
Project Title	Food Consumer Behavior Analytics Portal
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Develop an interactive, high-performance Tableau dashboard analyzing 50,000 food transaction records to identify consumer purchasing patterns across regions, age demographics, meal types, order frequency, and spending, and make the insights accessible through a Flask web portal.
Scope	Data loading and preprocessing; creation of the Age Group calculated field; development of 9 visualization sheets; assembly of a 3x3 interactive dashboard; City-based filtering; Tableau Public publishing; and Flask/HTML5 web embedding.
Problem Statement	
Description	Decision-makers and analysts face difficulty deriving actionable insights from a large food transaction dataset because raw transactional data lacks clear visual structure, demographic segmentation, interactive filtering, and a unified web-accessible dashboard.
Impact	The solution enables faster analysis, improved regional demand and inventory decisions, clearer demographic understanding, and more effective communication of consumer trends to business stakeholders.
Proposed Solution	
Approach	Load and verify the 50,000-row CSV dataset in Tableau; create calculated fields such as Age Group using IF/ELSEIF logic; build 9 analytical sheets; assemble them into a 3x3 dashboard with a global City filter; publish to Tableau Public; and embed the published dashboard in a Flask application using Tableau's JavaScript Embedding API.
Key Features	9-part 3x3 interactive dashboard; City-based global filtering; Age Group demographic segmentation; revenue and order analysis by meal type and city; Tableau Public deployment; responsive Flask web integration; full-width and full-height dashboard rendering.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7, 4+ cores recommended
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD; approximately 50 MB dataset footprint
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	pandas, jinja2
Development Environment	IDE, version control	VS Code, Tableau Desktop, Git
Data		
Data	Source, size, format	50,000 food transaction records in CSV format